

L14 Introduction to Markets

CS 280 Algorithmic Game Theory

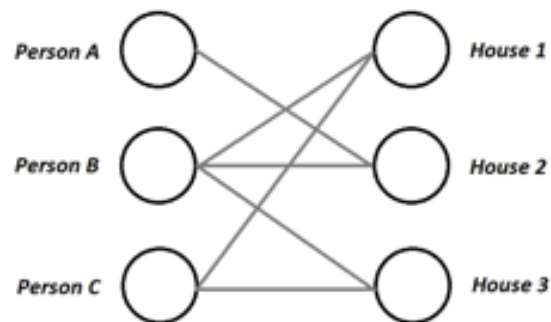
Ioannis Panageas



Food Markets



Stock Markets



Matching Markets

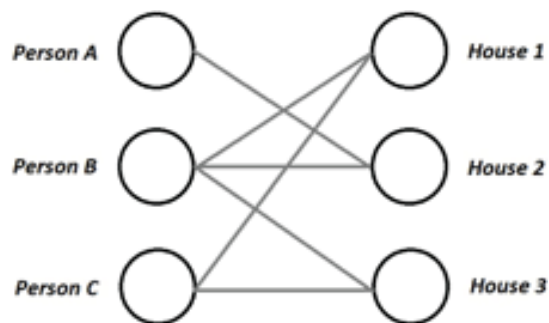
Driven by a rule: Supply meets demand!



Food Markets



Stock Markets



Matching Markets

Definitions

Definition (Market). *A market consists of:*

- *A set $\mathcal{B}$ of n buyers/traders.*
- *A set $\mathcal{G}$ of m goods.*
- *Each buyer i has e_i amount of \$. W.l.o.g assume $e_i = 1$.*
- *b_j denotes the amount of each good. W.l.o.g $b_j = 1$.*
- *u_{ij} denotes the utility derived by i on obtaining a unit amount of good of j .*
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Definition (Fisher Market). *A market so that the utilities are linear:*

Let x_{ij} be the amount of units buyer i gets of good j then

$$u_i = \sum_{j \in \mathcal{G}} x_{ij} u_{ij}.$$

Definitions

Definition (**Market clearance**). A vector of price (x^*, p^*) is called *market equilibrium* if for given prices p^* , *each buyer is assigned an optimal basket of goods* relative the prices and buyer's budget and there is *no surplus or deficiency* of any of the goods

Goal: Compute allocations and prices in polynomial time!

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$$\text{s.t } \sum_{j=1}^m p_j x_{ij} \leq 1$$

$$x_i \geq 0$$

Budget constraint.

Eisenberg-Gale Convex Program

Given an arbitrary vector of prices $p \geq 0$, from each buyer's i perspective:

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From the perspective of good j :

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Demand for good j .

Supply for good j .

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Can we find (x, p) s.t all are satisfied simultaneously?

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Consider the following **convex** program:

$$\begin{aligned} \max \quad & \sum_{j=1}^n \ln u_i \\ \text{s.t.} \quad & u_i = \sum_{j=1}^m u_{ij} x_{ij} \text{ for all buyers } i \in \mathcal{B}, \\ & \sum_{i=1}^n x_{ij} \leq 1 \text{ for all goods } j \in \mathcal{G}, \\ & x_{ij} \geq 0 \text{ for all } i \in \mathcal{B}, j \in \mathcal{G}. \end{aligned}$$

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Is x^* an equilibrium? What are the prices?

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x^* satisfies the KKT conditions.

KKT are first-order conditions for constrained Optimization

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KKT are first-order conditions for constrained Optimization

$$L(x, p) = \underbrace{\sum_{j=1}^n \ln u_i}_{\text{objective}} - \sum_{j=1}^m \underbrace{p_j \left(\sum_{i=1}^n x_{ij} - 1 \right)}_{\text{constraint for good } j}$$

Remark: Langrangian involves **objective and constraints**!

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KKT conditions: x are **primal** variables, p are **dual** variables.

Primal feasibility:

$x_{ij} \geq 0$ for all $i \in \mathcal{B}$, $j \in \mathcal{G}$.

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$$\frac{\partial L(x, p)}{\partial x_{ij}} = \frac{u_{ij}}{u_i} - p_j = 0 \text{ if } x_{ij} > 0.$$

$$\frac{\partial L(x, p)}{\partial x_{ij}} = \frac{u_{ij}}{u_i} - p_j \leq 0 \text{ if } x_{ij} = 0.$$

$$\frac{\partial L(x, p)}{\partial p_j} = 1 - \sum_{i=1}^n x_{ij} = 0 \text{ if } p_j > 0.$$

$$\frac{\partial L(x, p)}{\partial p_j} = 1 - \sum_{i=1}^n x_{ij} \geq 0 \text{ if } p_j = 0.$$

Complementary Slackness

Eisenberg-Gale Convex Program

Let (x^*, p^*) satisfy the **KKT conditions**. Then (x^*, p^*) solves

$$\min_{p \geq 0} \max_{x \geq 0} L(x, p) = \max_{x \geq 0} \min_{p \geq 0} L(x, p) \text{ since it is } \textit{convex} - \textit{concave},$$

where $L(x, p) = \sum_{j=1}^n \ln u_i - \sum_{j=1}^m p_j (\sum_{i=1}^n x_{ij} - 1)$.

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- *The set of equilibrium allocations is convex.*
- *Equilibrium utilities and prices are unique.*
- *If all u_{ij} 's are rational then allocations and prices are rational.*

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By KKT we have there exists buyer i so that $u_{ij} > 0$. We conclude from KKT

$$p_j^* \geq \frac{u_{ij}}{\sum_{j'=1}^m u_{ij'} x_{ij'}^*} > 0.$$

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Proof cont. Let x^* be an optimum of EG program and let p^* be the dual variables so that (x^*, p^*) satisfy the KKT constraints. We shall show that (x^*, p^*) is a **market equilibrium**.

1) We showed that $p_j^* > 0$ for all $j \in \mathcal{G}$.

Positive prices $\Rightarrow$

By complementary slackness we have $\sum_{i=1}^n x_{ij}^* = 1$.

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Using KKT conditions for fixed buyer i we also have for $x_{ij}^* > 0$

$$\frac{u_{ij}}{\sum_{j'=1}^m x_{ij'}^* u_{ij'}} = p_j^* \Rightarrow \frac{u_{ij} x_{ij}^*}{\sum_{j'=1}^m x_{ij'}^* u_{ij'}} = x_{ij}^* p_j^*$$

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Summing over all goods $j \in \mathcal{G}$ the above we have

$$1 = \frac{\sum_{j=1}^m u_{ij} x_{ij}^*}{\sum_{j'=1}^m x_{ij'}^* u_{ij'}} = \sum_{j=1}^m x_{ij}^* p_j^*$$

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Buyers spent all their money

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Hence (x^*, p^*) is a **market equilibrium**. Since EG is a convex program, the set x^* of optimal solutions to EG is a convex set.

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By doing the transformation $q_j = \frac{1}{p_j}$ the prices should satisfy a linear system (by KKT conditions) with **rational coefficients**.

Other utility functions

CES (Constant elasticity of substitution) utility functions:

$$u_i(x) = \left(\sum_{j=1}^m u_{ij} x_{ij}^{\rho} \right)^{\frac{1}{\rho}}, \text{ for } -\infty < \rho \leq 1.$$

Remark:

- $u_i(x)$ is **concave** function.
- If $u_{ij} = 0$, then the corresponding term in the utility function is **always 0**.
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$\rho = 1$ $\longrightarrow$ Linear utility form

$\rho \rightarrow -\infty$ $\longrightarrow$ Leontief utility form

$\rho \rightarrow 0$ $\longrightarrow$ Cobb-Douglas form

Proportional Response Dynamics

Market dynamics:

Each **time step** the buyers face the **same** market parameters, (**goods, budget constraint, utility function**) while the buyers make their **bidding decisions** according to the **previous** market actions

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Notation:

- $b_{ij}^{(t)}$ the **bid** of buyer i for good j at time t .
- $p_j^{(t)} = \sum_{i \in \mathcal{B}} b_{ij}^{(t)}$ **price** for good j .
- **Allocation** $x_{ij}^{(t)} = \frac{b_{ij}^{(t)}}{p_j^{(t)}}$.
- **Utility** of agent i from good j is $u_{ij}^{(t)} = x_{ij}^{(t)} w_{ij}$.
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Theorem (Convergence). *The proportional response dynamics converges to a market equilibrium in the Fisher market with linear utility functions.*

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Remark:

- The convergence result holds for **CES utilities** with a different rate.
- Similar rate to Multiplicative Weights Method (**not a coincidence**).

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The potential function will be (show it is decreasing)

$$\Phi^{(t)} = \sum_{i \in \mathcal{B}} \text{KL}(b_i^* || b_i^{(t)}).$$

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Remark:

- **KL divergence** $\text{KL}(x || y) = \sum x_i \log \frac{x_i}{y_i}$ for distributions x, y .
- $\text{KL}(x || y) \geq 0$, **pseudo-distance, symmetry not satisfied**.